

# Improved Edge Detection Model Based on HED

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**Abstract**—Influenced by the HED network structure, how to further solve the image multi-scale multi-level representation problem and optimize the detection of the overall image. We propose an end-to-end network structure. While enhancing the robustness of backbone network, the BN layer is used to solve the problem of network gradient dispersion, the attention mechanism is added to strengthen the role of deep supervision and final fusion module. The network can learn related semantic features autonomously, obtain richer image information, and solve the problem that HED cannot fully extract features, and high-level and low-level information are simply fused together. Considering the modification on the basis of the HED network structure, the BN layer and the SE structure are added, and the manner of downsampling is modified. The experimental results show that the edge effect from the BSDS300 data set is good, and the running speed is better than the HED model.

**Keywords**—Attention Mechanism; Deep Learning; Edge Detection; Convolution Neural Network; Image processing

## I. INTRODUCTION

Object edge detection is an important part of computer vision. In 2014 IGSVRC VGG [7], in order to solve the degradation caused by the number of network layers, K He [6] et al proposed a skip connection to solve this problem, in order to be able to identify a specific part of the picture, Jonathan Long [1] in CNN On the basis of the elimination of the fully connected layer, a full convolutional neural network is formed, there is only one loss function, the output prediction generated is not multi-scale. Xie [2] proposed a deep supervisory layer on the basis of FCN to maintain 5 losses functions at the same time; Hou [8] proposed to use short connection to strengthen the relationship between high and low layers, while maintaining six loss functions; Liu [5] makes additional convolutions on each backbone network similar to deep separable convolutions, and finally each of the results of each side output is integrated, and also maintain six loss functions; Dai [4] and S.Ren [9] et divide the Region-of-Interest respectively combined with FCN or VGG, K He[6] pools the latter region of interest (ROI) into two convolution groups, sacrificing the speed of operation, but improving the accuracy; Fei Wang [11] proposed a Use of attention mechanisms and combined with [6], in CIFAR-10 And the CIFAR-100 dataset performed better than the Resnet-200. In [3] SE structure can weight feature channels, highlight important channels, and dilute channels that are of little use.

More and more deep learning is reintegrated into the concept of machine learning and can achieve good results. In this paper, a model combining VGG [7] and SE [3] modules is proposed, and the part of backbone network is modified on this basis. The general CNN network (including VGG) can increase the receptive field by downsampling to capture the overall pattern of the image to express the picture information, and the SE structure can take advantage of the relationship between the channels by weighting the channel intelligently. The operation is to highlight the importance of each channel, and the SE structure itself has good compatibility with other network structures. The addition of the BN layer can effectively solve the problem of gradient disappearance or dispersion.

## II. RELATED WORK

In the past few decades, the recognition of objects has only been discerned by the human eye, which is extremely labor intensive. Until the emergence of image processing, Sobel [22] template provides the most advanced solution at the time for edge and object detection. And become a classic textbook-like algorithm today, and Sobel templates can get good results in filtering out smooth noise. However, the computing power of computers in the last century is far from being able to process pictures on a large scale until the emergence of machine learning, including K-means algorithm [24], K-nearest neighbor algorithm [25], etc., and new image processing methods are integrated into people's vision. Among the visual fields, the famous method is the support vector machine [23]. In 2016, the rapid rise of artificial intelligence led to the entire information age, in which traditional machine learning faced problems—depending on manual extraction of too many features, and different extraction criteria, different parameter settings and weight assignments — It is difficult to have a longer-term development. For image processing and natural language processing, convolutional neural networks can use their multi-scale and multi-level network learning to control the information between the whole and the part for research purposes.

The object detection in image processing lies in solving the main contour by extracting the target. The main algorithms include Sobel [22], etc. In order to reduce the difficulty of manual extraction, the basic physical features of the target are extracted in the shallow layer by training the neural network

model. Such as simple contours, etc., and as the number of layers in the network deepens, it is possible to extract high-dimensional and abstract information, such as eyes, in such a way as to extract the semantic features of the image. In the ILSVRC competition, the VGG [7] model was proposed to improve the CNN to increase the receptive field by using the 3x3 small convolution kernel, and to use the small pool to complete the downsampling. In order to obtain information of different scales, Google proposed the Inception [17] network. A large number of experiments show that the deepening of the network layer can effectively improve the expressive ability of the network itself, but with the deepening of the deep network, it faces the side effects such as gradient disappearance and dispersion, causing network degradation, the Dropout [12] and BN [16] layers can effectively solve this problem. In 2015, the ResNet[6] network came out. In the same year, the HED [2] network also appeared. The former succeeded in avoiding the problem of gradient disappearing while deepening the network. The latter was deeply affected by the FCN [1] network, canceling the general fully connected layers, the basic network uses the front convolution and pooling layers of VGG [7]. At the end of each stage, the last feature map is taken for upsampling, and finally the feature map is merged. Both RCF [5] and DSS [8] modify based on HED in different ways. RCF uses a fusion strategy for all layers in each stage. DSS deepens a pooling layer in depth. The line is upsampled and fused with the previous particular layer, thus maintaining 7 loss functions simultaneously. Despite this, convolutional neural networks still have a lot of room for improvement in image semantic feature extraction, multi-scale and multi-level

### III. EDGE DETECTION MODEL

#### A. HED NETWORK

HED refers to the FCN [1] network, which replicates end-to-end edge detection. The performance of HED can achieve one of the best results at present by training on existing data sets. As shown in Figure 1, the HED [2] network structure selects VGG16 on the backbone network, and after discarding its last fully connected layers and the last pooled layer, it can be divided into five stages. HED implements the edge on this basis. Feature extraction and image semantic processing analysis, the result extracted by each Stage then throughs 1x1 convolved layer, followed by upsampling, the result is compared with the Ground Truth value. After calculating the loss, parameters of the neural network will be trained by BP algorithm, and End-to-end edge detection is achieved because all layers are convolutional. Moreover, HED has more side output, and these side outputs after deep supervision are better than the result obtained by pure FCN [1]. In edge detection experiments, it is often more reasonable to calculate the integration of multiple predictions than a single loss function or output to measure in the quality of the entire network.

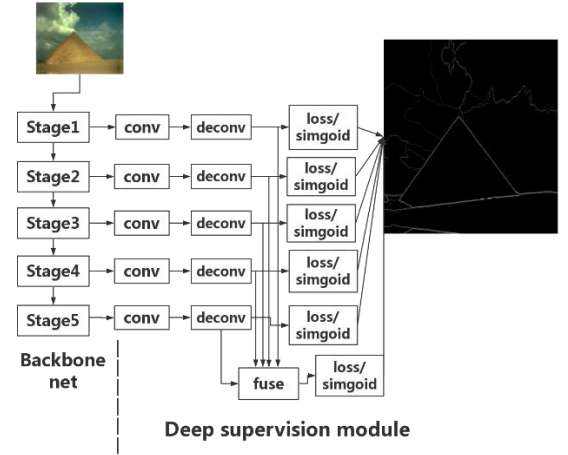


Figure 1: The detail of HED network

#### B. BACKBONE NETWORK MOIDULE

This proposed model refers to the HED [2] network, similar to the HED network structure, this proposed model also retains the backbone network, deep supervision and feature fusion module. However, improvements have been made in the experimental tests in each module. The specific structure is shown in Figure 2.

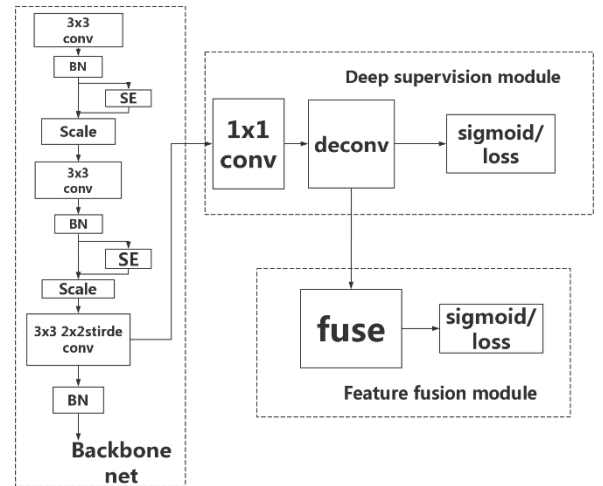


Figure 2 :The details of the proposed network

##### 1) Backbone network model

In the case where the gradient of the network layer is too deep and the gradient is dispersed, the general network will have better network performance as the number of layers is deepened. At the beginning, we considered using the Resnet-101 [6] network as the main network, but the parameters are too many to control. Considering the improvement measures afterwards, the memory capacity is insufficient, and the effect of each stage is not as good as VGG. Based on these two points, the model considers the VGG19 model, which adds a layer to each of Stages 3, 4, and 5.

In [2], considering the characteristics of hierarchical learning, multi-scale learning can be considered to be expressed in the “inside” of the neural network by using the downsampling layer and expanding the receptive field. The obtained feature map is multi-scale. It can also be thought of as being passed in “outside”, that is, passing images of different scales. In fact, these two methods are similar. In “internal”, the common way to expand the feeling is through pooling and convolution, where for the image data, the largest pooling layer can intuitively control the sudden change of image data, while reducing the subsequent convolution template processing. For each channel's data, taking into account the translational variability in [4] target detection, then there is also a certain pattern in the feature map. If the average pool is used, the maximum pooling can get more information. However, as the network becomes deeper, the input of the target at different locations may result in the inability to perceive the existence of the target after the next few rounds of pooling. Combined with the downsampling method in Resnet [6], we consider replacing the largest pooled layer in VGG with a convolution layer with 2 steps.

The main solution to the gradient dispersion that may be caused by network deepening can be through Dropout and BN. The Dropout layer is mediated by randomly rounding off parameters so that the features learned by each neuron are independent of each other, avoiding the inherent dependence between neurons; while the BN layer normalizes the gradient by image data normalization. We consider adding the BN structure to the structure of this article.

## 2) Combined with the backbone network of the Squeeze-and-Excitation Network

Although we simply use a variety of downsampling methods, nonlinear functions and BN layers, it is enough to capture the global receptive field and feature patterns, but through the SE structure, Hu et al. shift the focus to the relationship between different channels, not just in the spatial relationship of features, multi-scale and multi-level issues. As shown in Figure 3, each channel in the SE structure, the size of the  $H \times W$  feature map  $u$ , is first globally pooled into a real number  $a_i$ , the formula is as follows:

$$a_i = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j) \quad (1)$$

These real numbers  $a_i$  with global receptive fields are connected together to form a new feature map of  $1 \times 1 \times C$ , where  $C$  is the number of channels of the initial feature map, and then the weight  $s$  of the initial feature map is formed by using the learnable parameter  $w_i$  corresponding to  $a_i$ . Its formula is as follows:

$$s = \sigma(w_2, \delta(w_1, a)) \quad (2)$$

Finally, we weight the  $s$  and the initial feature map, the formula is as follows:

$$\tilde{x}_c = F_{scale}(u_c, s_c) = s_c u_c \quad (3)$$

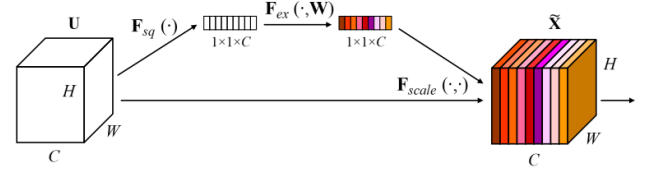


Figure 3 : Detail of Squeeze-and-Excitation network [3]

Through this weighted SE structure, the importance of the image channel can be discriminated, and the channel containing important information can be emphasized, and the influence of the useless information channel on the performance of the subsequent network can be reduced.

We consider SE structuring the feature map after each layer has been convolved. It has been experimentally found that this structure can effectively improve the overall network performance. Helps to accurately identify the network for object recognition and effectively isolate unwanted information. The effect is shown in Figure 4. It can filter the feature maps obtained in the entire stage in the feature extraction of the first layer, emphasizing the useful feature maps and reducing the effect of the chaotic feature maps in the fusion phase.

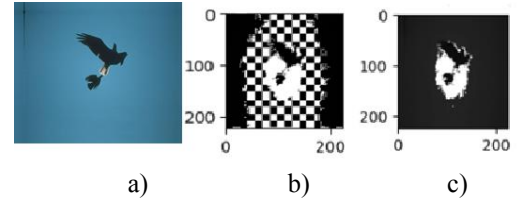


Figure 4 : The effects of SE network: a) is original picture; b) training result without SE; c) is training result with SE network

## 3) Deep supervision module

The neural network that has been used “deep but not wide” can only extract local information, but could be accurate through repeatedly refining the feature information. Although the appropriate depth of the network does have a good effect in recognition and classification, there are too many training parameters caused many methods unfeasible due to running conditions is costly. It is also easy to cause the processor occupancy rate to be too high to be further improved. Furthermore, the experimenters cannot easily understand what the abstract information extracted from the different depth of the neural network. By training independently on the “width”, that is, by compressing the feature by  $1 \times 1$  convolution and then using up-sampling to achieve upsampling, the multi-scale information can be better fused, after passing through the SE [3] structure, at the end fusion module can better show the effect. And since the five stages each match a loss function, they remain independent of each other, do not affect each other's results, and reduce the information of the redundant fusion in the final fusion module.

#### 4) Feature fusion module

Each stage gets a feature map after downsampling and feature extraction, simply extracting the last feature map, and then using deconvolution to achieve end-to-end effect is not ideal, considering FCN [1] and HED [2] The practice, combined with the feature maps of different downsampling levels, and then connected together, the recognition effect is significantly better than the former. The shallow neural network extracts all the low-level simple edge features, and the extraction of semantic features is not enough. After the deep feature maps undergo a large amount of convolution extraction, the obtained feature maps have a large number of semantic features, but they are discarded. Basic edge features. Therefore, it is a reasonable thing to combine the features of depth. By adding the SE structure to the last layer, the network can discriminate the focus of the fusion and realize the preservation of the edge and semantic features of the image object.

Considering that the feature fusion module of the model in this paper adds nonlinear elements, the edge effect obtained by the experiment is worse than that obtained by adding no nonlinear elements. Therefore, no nonlinear elements are added in the feature fusion module of the model.

#### IV. LOSS FUNCTION

Considering that the result of this paper is the edge feature extraction of the original image, then each pixel point needs to be classified, and the classification of the pixel points is regarded as a two-classification problem using a simple nonlinear function pair, then each side output is Can be seen as a classifier, the score of each pixel is a probability value. Each classifier corresponds to a set of deep supervision module parameters  $w = (w_1^{(1)} \dots, w^{(5)})$ , and all the parameters on the backbone network are unified into  $W$ , and the loss function of the side output is And can be identified as follows:

$$L_{side}(W, w) = \sum_{i=1}^5 \alpha_i l_{side}^{(i)}(W, w^{(i)}) \quad (4)$$

Among them,  $l_{side}^{(i)}$  is used to measure the degree of loss in the second classification of the last pixel. The final feature fusion module will also suffer losses, the function is as follows:

$$L_{fuse}(W, w, x_{fuse}) = l_{fuse}(W, w, x_{fuse}) \quad (5)$$

Where  $x_{fuse}$  is the probability value calculated by using the side output and then calculated by the sigmoid function. The overall loss function is the loss function after each side loss function and feature fusion:

$$L = L_{side}(W, w) + L_{fuse}(W, w, x_{fuse}) \quad (6)$$

Considering that there are many non-edge pixels in a graph than edge pixels, we add weights to the general cross-entropy calculation expression, assuming  $\gamma = \frac{|Y^+|}{|Y^+| + |Y^-|}$ . Then  $1 - \gamma = \frac{|Y^-|}{|Y^+| + |Y^-|}$ ,  $|Y^+|$  represents the number of feature pixel points, and  $|Y^-|$  represents the number of non-feature pixel

points For this phenomenon, the improved cross-entropy formula is as follows:

$$l_{side}^{(i)}(W, w^{(i)}) = -\gamma \sum_{y_j \in Y^+} \log S(y_i = 1|x; W, w^{(i)}) - (1 - \gamma) \sum_{y_j \in Y^-} \log S(y_i = 0|x; W, w^{(i)}) \quad (7)$$

The S function is a sigmoid function, which is responsible for converting the last upsampled feature map into a probability map,  $x_{fuse} = S(\sum_{i=1}^5 k_i \cdot \tilde{x}_{side}^{(i)})$ ,  $\tilde{x}_{side}^{(i)}$  represents the final feature map for each side.

The threshold method is used in the model of this paper [5]. First, the pixel values are normalized. Considering that the value greater than  $\eta$  is regarded as the edge point, considering that the possible label map is completed by multiple people, and each person has different understanding of the image feature points, and will be between 0 and  $\eta$ . Values are excluded as dispute points and are not considered. In this model, the  $\eta$  value is set to 0.5 and the initial learning rate is 1e-6.

#### V. COMPARED WITH HED

The difference between this model and HED is reflected in the deepening of the backbone network, that is, the deepening of the 3-layer network structure; the downsampling method is no longer implemented by the pooling of 2x2 steps in the original model, and the 2x2 step is used instead. Convolutional layer; insert SE structure modules in each layer of each backbone network to help the network understand the relationship between different channels; to prevent the occurrence of gradient dispersion and join the BN layer. Through the improvement of the above points, the performance of this model is better than the edge extraction effect of HED on the clear picture, and the training time is greatly improved.

#### VI. EXPERIMENT

The experimental configuration used in this model is as follows:

Table 1 : Configuration of experimental details

| Name                     | Related Configure |
|--------------------------|-------------------|
| CPU                      | I5-8600k          |
| Memory                   | 16G               |
| GPU                      | RTX2070           |
| GPU acceleration library | CUDA, CUDNN       |
| DATA processing          | Python, Keras     |
| Deep learning framework  | Keras             |

#### A. IMAGE ENHANCEMENT

The model performs random angle rotation, expansion and reduction, and cropping on the underlying BSDS300 dataset, and the dataset is enhanced to 3000 sheets and unified to a size of 224x224.

Table 2 : Detailed number on the receptive field, stride, channel numbers of the proposed model. C1-2etc respectively connect side output layer. C1-downetc means specific downsampling.

| Layer   | C1_2    | C1_down | C2_2    | C2_down | C3_4 |
|---------|---------|---------|---------|---------|------|
| rf size | 5       | 6       | 14      | 16      | 40   |
| Stride  | 1       | 2       | 2       | 4       | 4    |
| channel | 64      | 128     | 128     | 256     | 256  |
| Layer   | C3_down | C4_4    | C4_down | C5_4    |      |
| rf size | 44      | 92      | 100     | 192     |      |
| Stride  | 8       | 8       | 16      | 16      |      |
| channel | 512     | 512     | 512     | 512     |      |

In this paper, the RTX2070 graphics training model is used. In order to demonstrate the improvement of the model, the HED [2] model is trained with the same data enhancement set. The final edge effect of the fusion is shown in Figure 5. Considering the improvement over traditional edge detection, this group of experiments added the operation of Sobel template. From left to right, the original image is tested, the middle is the edge detection image obtained by the model, and the far right is the edge detection image obtained by HED under the same computing power. It can be clearly discerned from the figure that after the backbone model modified and the downsampling methods are replaced, the effect obtained is improved compared with the HED model, but both performances are not good when process on pictures with complicated texture. The proposed model has made some progress in the extraction of semantic features, and can clearly distinguish the range of green space (fourth group comparison), while the HED model divides the path and the distant sky into one edge (the fifth group comparison), and Too many extraneous points are extracted as edge points, and the resulting edges are blurred. The final Sobel template can get better results on the edge extraction of birds and airplanes, but in the face of the more complex green space, the edge results on the polar bears are not satisfactory, but the results in the extraction ship's contour are better, but because Lack of ground truth tag comparison, as well as semantic analysis of neural networks, extracts inconsequential edges, such as the rippled edges of water.

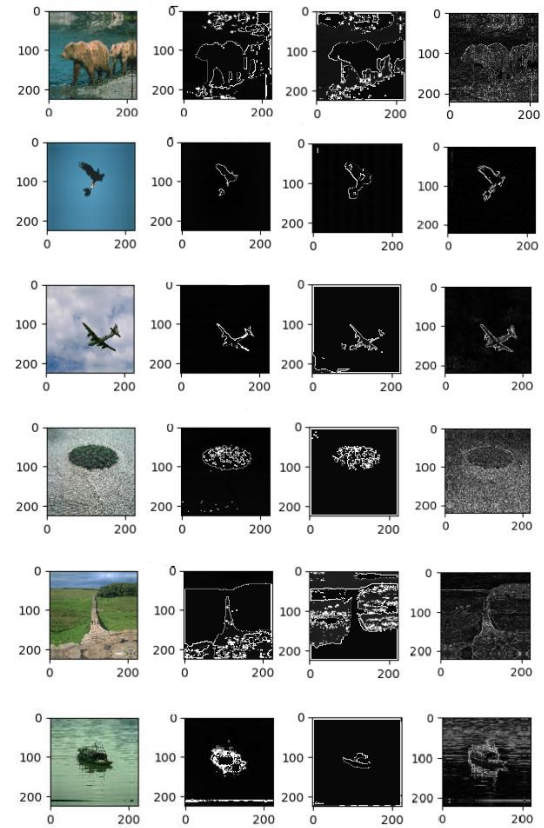


Figure 5: Final effect comparison chart. The left-most part is the original image, the second group is the edge detection image of this model, the third group is the edge effect image of HED after the same training with this model, and the last group is the result of Sobel template.

#### B. EXPERIMENTAL PERFORMANCE COMPARISON

As can be seen from Table 3 and Figure 5, although the model has slightly improved compared with the HED model in the AP index, it greatly improves the training speed.

Table 3: performance comparison

| network          | Time consuming/ms | AP    |
|------------------|-------------------|-------|
| HED              | 400               | 0.833 |
| Proposed Network | 139               | 0.837 |

#### VII. CONCLUSION

This article provides a new edge detection model. Drawing on the core ideas of HED, SE and Resnet. The network has improved the semantic analysis and feature extraction capabilities of images. Starting from FCN, in order to achieve end-to-end methods, more and more neural networks are no longer easy to consider fully connected layer. This model performs well in BSDS300 dataset which generates better edge detection maps than HED. However, the effect of this model on the image extraction features with more lines is still

not ideal, some novel and effective feature extraction measures can be improved in the future.

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